HL ONLY - Due Feb 02nd

EXERCISE 1

This question asks you to explore self-composite linear functions, of the form $f(x) = mx+c$, for varying values of m.

A function composed with itself is called a self-composite function.

For a function f, the function composition with itself is given by $(f \circ f)(x) = f(f(x))$.

Let f^n denote the nth composition of f with itself where

$$f^n(x) = (f \circ f \circ \cdots \circ f)(x) \qquad (n \text{ times}).$$

Hence, for example,

$$f^2(x) = (f \circ f)(x) \quad \text{and} \quad f^3(x) = (f \circ f \circ f)(x) = f(f(f(x))).$$

Consider the linear function $f(x) = mx + c$, where $x \in \mathbb{R}$ and $m, c \in \mathbb{R}$.

(a) Show that

(a.i)

$$f^2(x) = m^2x + c(1 + m);$$

[3 marks]

(a.ii)

$$f^3(x) = m^3x + c(1 + m + m^2).$$

[2 marks]

(b)

(b.i) Write down an expression for $f^4(x)$. [1 mark]

(b.ii) Suggest a similar expression for $f^n(x)$, $n \in \mathbb{Z}^+$. [2 marks]

(b.iii) By using your expression from part (b.ii), or otherwise, find an expression in terms of n for $f^n(x)$ when $m = 1$. [3 marks]

Questions below are optional

Consider

$$f^n(x) = m^n x + c\left(\frac{1-m^n}{1-m}\right) \quad \text{where } -1 < m < 1.$$

(d) As $n \to \infty$, the family of graphs $y = f^n(x)$ approaches the graph of a straight line, L. Determine the equation of L, giving your answer in terms of c and m. [4 marks]

Consider

$$f^n(x) = m^n x + c\left(\frac{1-m^n}{1-m}\right) \quad \text{where } m = -1.$$

(e)

(e.i) Show that $f^n(x) = -x + c$ when n is odd. [2 marks]

(e.ii) Find an expression for $f^n(x)$ when n is even. [2 marks]